

Genome Exploration II

Basic Genome Structure and Sequence Exploration Using Galaxy

Continuation Assignment • usegalaxy.org • Individual Work

Name	Batigulao, Jehiah Bless T.
Course and Section	BIO 300 - B
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Instructors	Lilibeth A. Bucol (on going PhD)

Part 1.

Species name	<i>Miniopterus schreibersii</i>
NCBI Assembly Accession	GCA_964146895.2
Assembly level (if known)	Scaffold
FASTA filename in Galaxy	<i>Miniopterus schreibersii</i> _original_genome.fna.gz.gz
Genome Source/Database	NCBI
Approximate file size	544Mb

The screenshot displays the Galaxy web interface. The central panel shows the dataset '1: Miniopterus_schreibersii_original_genome.fna.gz.gz' with 186 sequences, a FASTA format, and a size of 570.1 MB. A warning message indicates that the dataset is large and only the first 100 KB is shown. The preview displays a portion of the genome assembly for chromosome 1, starting with 'TTCTCtctccccctccccctctctgtgaatcaataaaacacattaaatataatataacagtcagaaagggtg'. The right sidebar shows the dataset's history, including its upload from a FASTA file, and its storage details, such as an average scaffold length of 9,549,438.40 and a total size of 4.32 GB.

Figure 1. *Miniopterus schreibersii* genome dataset in Galaxy History

Part 2.

Metric	Values	Interpretation
Total Assembly Length	1,850,590,334	The assembly is about 1.85 Gb, representing the total length of the assembled genome.
Number of sequences/scaffolds	186	The genome is made up of 186 scaffolds. A lower number of scaffolds generally indicates a more contiguous assembly.
Minimum Sequence Length	13,000 bp	The shortest scaffold is 13 kb long.
Maximum Sequence Length	202, 313, 579 bp	The longest scaffold is about 202.3 Mb, showing that the assembly contains very long sequences.
Mean/Average Sequence Length	9,949,410.40 bp	The scaffolds have an average length of about 9.95 Mb.
N50	87,795,433 bp	Half of the assembly is contained in scaffolds that are 87.8 Mb or longer, indicating relatively good contiguity.
L50	8	Only 8 of the longest scaffolds are needed to contain at least half of the assembly, showing that much of the genome is concentrated in a few long scaffolds.
GC Content (%)	42.51	About 42.5% of the assembly is composed of G and C bases.

Column 1	Column 2
# scaffolds	186
Total scaffold length	1,850,590,334
Average scaffold length	9,949,410.40
Scaffold N50	87,795,433
Scaffold auN	100,270,149.34
Scaffold L50	8
Largest scaffold	202,313,579
Smallest scaffold	13,000
# contigs	326
Total contig length	1,850,562,334
Average contig length	5,676,571.58
Contig N50	28,014,000
Contig auN	32,192,266.13
Contig L50	21
Largest contig	83,366,000
Smallest contig	13,000
# gaps in scaffolds	140

Figure 2. *Miniopertus schreibersi* Statistics output using gfastats in Galaxy History.

Part 3.

Rank	Sequence ID	Length (bp)
1	NC_138101.1	202313579
2	NC_138102.1	187805147
3	NC_138103.1	105355512
4	NC_138105.1	102193545
5	NC_138106.1	92835542

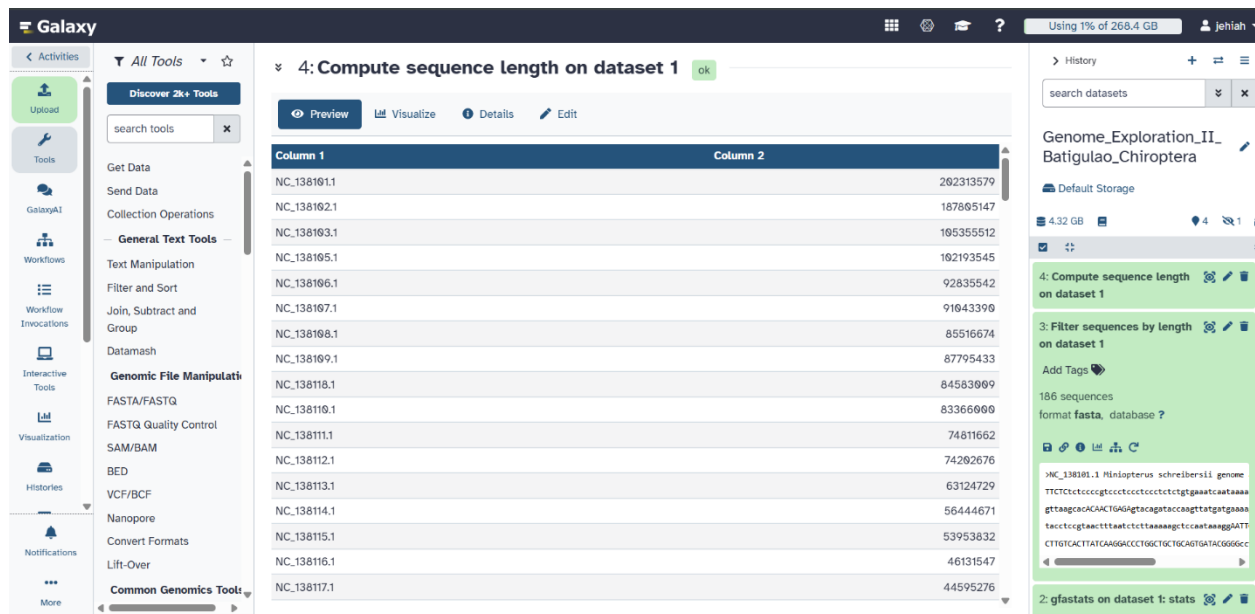


Figure 4. *Miniopterus schreibersii* Sequence Length Output using Compute Sequence Length in Galaxy History

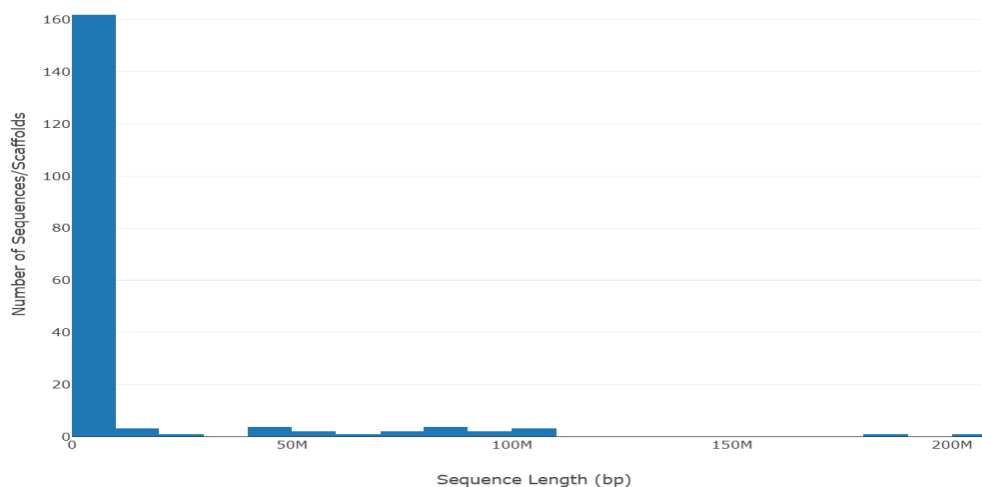


Figure 5. *Miniopterus schreibersii* Sequence Length Distribution using Histogram of a Numeric Column in Galaxy History. The graph shows that most scaffolds are relatively short, while only a few scaffolds are very long, reaching over 200 Mb. This indicates a mixture of sequence lengths, with many shorter scaffolds and several large scaffolds.

Part 4.

Metric	Original Genome	After ≥ 10 kb filter
Total length	1,850,590,334 bp	1,850,590,334
Number of sequences	186	186
Maximum length	202,313,579 bp	202,313,579
N50	87,795,433 bp	87,795,433
GC Content	42.51	42.51

The screenshot shows the Galaxy web interface. The main panel displays the output of the 'Filter sequences by length' tool, showing a large block of genomic data for Miniopterus schreibersii. The right sidebar shows the history of the workflow, including the 'gfastats on dataset 5: stats' tool output.

Figure 6. *Miniopterus schreibersii* Filtered 10kb Sequences Output using Filter Sequences by Length in Galaxy History

The screenshot shows the Galaxy web interface. The main panel displays the output of the 'gfastats on dataset 5: stats' tool, showing a table of assembly statistics for the filtered 10kb assembly. The right sidebar shows the history of the workflow, including the 'gfastats on dataset 5: stats' tool output.

Column 1	Column 2
# scaffolds	186
Total scaffold length	1,850,590,334
Average scaffold length	9,949,410.40
Scaffold N50	87,795,433
Scaffold auN	169,279,149.34
Scaffold L50	8
Largest scaffold	262,313,579
Smallest scaffold	13,969
# contigs	326
Total contig length	1,850,562,334
Average contig length	5,676,571.58
Contig N50	28,014,069
Contig auN	32,192,266.13
Contig L50	21
Largest contig	83,366,099
Smallest contig	13,969
# gaps in scaffolds	149

Figure 7. *Miniopterus schreibersii* Filtered 10kb Assembly Statistics Output rerun using gfastats in Galaxy History

- How many sequences were removed by the 10 kb filter?

No sequences were removed because the shortest scaffold was already 13,000 bp, which is above the 10 kb cutoff.

- Did the total assembly length decrease greatly or only slightly?

The total assembly length did not decrease because all sequences were already at least 10 kb long.

- Did N50 change? Why do you think it changed or stayed similar?

N50 stayed the same at 87,795,433 bp because no sequences were removed by the filter.

- What does this tell you about the contribution of short sequences to your assembly?

It shows that sequences shorter than 10 kb did not contribute to this assembly because none were present.

Part 5.

Metric	Values
Selected sequence	NW_027669791.1
Sequence length	190,730 bp
Minimum ORF size used	300 bp
Number of ORFs found	76
Longest ORF	97,671 bp

- How many ORFs were found in the selected sequence/region?

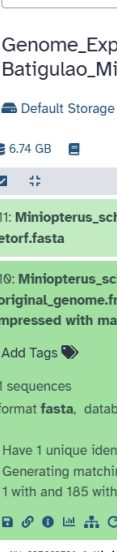
A total of 76 ORFs were found in the selected sequence.

- What was the longest ORF?

The longest ORF was 97,671 bp long.

- Why should you NOT report every ORF as a real gene?

ORFs only indicate sequences that have the potential to encode proteins. They are not automatically real genes because additional evidence is needed to confirm their biological function.



History

search datasets

Genome_Exploration_II_ Batigulao_Miniopterus...

Default Storage

6.74 GB

7 4 1 2

11: Miniopterus_schreibersii_g
etorf.fasta

11: Miniopterus_schreibersii
original_genome.fna.gz unco
mpressed with matched ID

Add Tags

1 sequences

format **fasta**, database ?

Have 1 unique identifiers from list
Generating matching FASTA file
1 with and 185 without specified IDs

1 2 3 4 5

```
>MW_027669791.1 Miniopterus schreibersii geno
ccttagcctcccaacataacccctaaccccttaggtccaatcgtaaccc
ctaaccccttaggtccaatcgtaacccctaaccccttaggtctctaacctta
aacccctaacggatcttagtccttagcctcccaacataacccctaac
```

[illegible]

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Part 6.

The genome assembly appears to be relatively contiguous rather than highly fragmented, based on the large scaffold and contig N50 values and the presence of several very long sequences. The scaffold N50 of about 87.8 Mb means that half of the assembled genome is contained in scaffolds at least this long, while the L50 of 8 indicates that only eight scaffolds are needed to account for half of the genome. The maximum scaffold length of about 202.3 Mb also shows that the assembly contains very large continuous regions. Although there are 326 contigs and 186 scaffolds, the relatively large N50 values suggest that many of these sequences do not greatly fragment the overall genome. The difference between the total genome length and the N50 values also indicates that many short sequences are present, but they contribute relatively little to the total genome size compared with the larger scaffolds. The genome has a GC content of 42.51%, which represents the proportion of guanine and cytosine bases in the assembled sequence. Finally, the ORF exercise demonstrated that the genome contains regions that can potentially encode proteins, while also showing how ORF prediction can identify possible coding regions based on sequence length and start/stop codons.